

The Dependence of Asteroid Rotation on Composition

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Motivation and Science Question

Asteroid spin states are shaped by collisions, YORP torques [1], and internal damping. Their rotation can therefore act as a population-scale probe of internal structure and material properties.

- ▶ A **rotational valley** separates faster and slower rotators in period–diameter space [2].
- ▶ Asteroid spin-rate distributions also separate into faster and slower rotator populations [3].
- ▶ The valley may be controlled by the balance between collisional excitation and internal damping.
- ▶ Spectral class provides a large-scale proxy for asteroid composition.

Science question

Does composition shift the rotational valley in period–diameter space?

S-complex

Silicate-rich
ordinary-chondrite-like

stronger / less dissipative

C-complex

Carbonaceous / primitive
often more porous

weaker / more dissipative

Hypothesis:

$$P_C^*(D) \neq P_S^*(D)$$

Data and Sample Selection

This work extends MP3C by combining asteroid physical properties with a best spectral class per object.

- ▶ Diameter, visible albedo, rotational period, and spectral class were compiled for each asteroid.
- ▶ Spectral classes were selected using a score based on method, wavelength coverage, taxonomy scheme, and probability.
- ▶ Periods were filtered using LCDB quality flags of at least 2–.
- ▶ Albedo cuts reduced contamination:

$$p_V \leq 0.12 \quad \text{for C-complex,} \quad p_V > 0.12 \quad \text{for S-complex.}$$

Population	Included classes	Final sample
S-complex	S, Sa, Sq, Sr, Sv	6049
C-complex	B, C, Cb, Cg, Cgh, Ch	2947

Method

The rotational valley was fitted in log period–log diameter space using the semi-supervised Support Vector Machine method of Zhou et al. (2025) [2], applied separately to each spectral complex.

$$\log P = k \log D + b \quad P^*(D) = B \left(\frac{D}{1 \text{ km}} \right)^k$$

- ▶ The fitted boundary traces the low-density valley between fast and slow rotators.
- ▶ Diameter uncertainties were tested using 1000 synthetic datasets.
- ▶ Robustness was tested using different spectral-score thresholds.

Robustness Checks

The compositional offset is not produced by a single filtering choice.

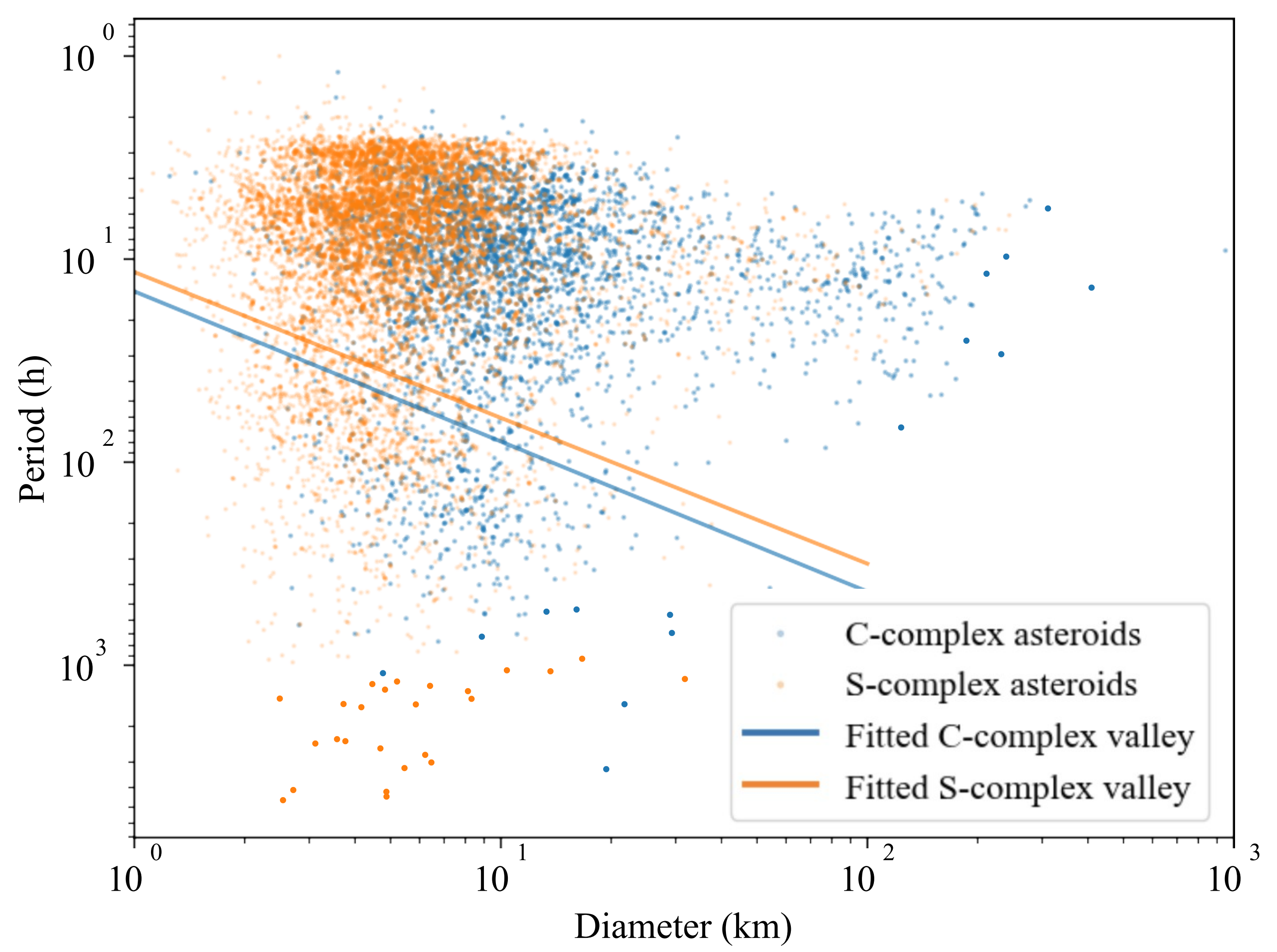
- ▶ Diameter uncertainties were tested using 1000 synthetic datasets.
- ▶ The valley was re-fitted using different spectral-score thresholds.
- ▶ The separation remains with increasingly strict spectral-class selection.

Key references

1. Bottke et al. (2006), *AREPS: The Yarkovsky and YORP Effects: Implications for Asteroid Dynamics*.
2. Zhou et al. (2025), *Nature Astronomy: Confined tumbling state as the origin of the excess of slowly rotating asteroids*.
3. Pravec et al. (2008), *Icarus: Spin rate distribution of small asteroids*.
4. Fienga et al. (2020), *MNRAS: Asteroid masses obtained with INPOP planetary ephemerides*.

Main Result

The fitted period–diameter valleys are offset between the two compositional populations.



Key result

The C-complex valley occurs at **longer rotation periods** than the S-complex valley.

Quantitative Comparison

Population	Valley relation	Interpretation
C-complex	$P_C^* = 14.4 D^{0.739}$	Longer-period valley
S-complex	$P_S^* = 11.6 D^{0.718}$	Shorter-period valley

For a given diameter, the fitted C-complex valley lies at longer periods than the S-complex valley:

$$P_C^*(D) > P_S^*(D).$$

This provides quantitative evidence that asteroid rotational evolution depends on composition.

Interpretation and Conclusions

- ▶ Collisions can excite non-principal-axis rotation.
- ▶ Internal friction damps excited rotation back toward principal-axis rotation.
- ▶ More dissipative bodies damp excited spin states more efficiently.
- ▶ Using representative densities for C- and S-complex asteroids [4], the valley offset implies a lower effective μQ for C-complex asteroids.
- ▶ The observed offset is consistent with C-complex asteroids being more porous, weaker, and more dissipative than S-complex asteroids.

$$(\mu Q)_C < (\mu Q)_S$$

Take-home message

Asteroid rotation depends on
size *and* composition.

Scan for the paper.

